

5. Let $\mathbf{u}_1 = (1, 1, 1)^T$, $\mathbf{u}_2 = (1, 2, 2)^T$, and $\mathbf{u}_3 = (2, 3, 4)^T$.

(a) Find the transition matrix corresponding to the change of basis from $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

(b) Find the coordinates of each of the following vectors with respect to the ordered basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$:

(i) $(3, 2, 5)^T$ (ii) $(1, 1, 2)^T$ (iii) $(2, 3, 2)^T$

(c)
$$E = \begin{bmatrix} | & | & | \\ \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$U = \begin{bmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

$$E \cdot [\mathbf{v}]_E = U \cdot [\mathbf{v}]_U$$

$$\Rightarrow U^{-1} \cdot I \cdot [\mathbf{v}]_E = U^{-1} \cdot U \cdot [\mathbf{v}]_U$$

$$\Rightarrow U^{-1} \cdot [\mathbf{v}]_E = [\mathbf{v}]_U$$

As we can see, U^{-1} is the transition matrix itself

$$[U | I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right] = [I | U^{-1}]$$

$$U^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

cb)

$$\begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} = U \cdot (V)_U$$

$$\Rightarrow (V)_U = U^{-1} \cdot \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$= 3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} + 5 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \#$$

ii.

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = U \cdot (V)_U$$

$$\Rightarrow (V)_U = U^{-1} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \#$$

$$\text{iii.} \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} = U \cdot (V)_U$$

$$\Rightarrow (V)_U = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \#$$

15. Let A be a 4×5 matrix and let U be the reduced row echelon form of A . If

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 1 \\ -3 \\ -2 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} -1 \\ 2 \\ 3 \\ 1 \end{bmatrix},$$

$$U = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) find a basis for $N(A)$.

(b) given that $\mathbf{x}_0$ is a solution to $A\mathbf{x} = \mathbf{b}$, where

$$\mathbf{b} = \begin{bmatrix} 0 \\ 5 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_0 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

(i) find all solutions to the system.

(ii) determine the remaining column vectors of A .

$$(a) \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & -2 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \mathbf{0}$$

$$x_3 = \alpha, \quad x_5 = \beta, \quad \begin{aligned} x_1 &= -2\alpha + \beta \\ x_2 &= -3\alpha + 2\beta \\ x_4 &= -5\beta \end{aligned}$$

$$\begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ -5 \\ 1 \end{bmatrix}, \quad N(A) = \left\{ \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ -5 \\ 1 \end{bmatrix} \right\}^*$$

(b) (i)

x_0 is one of the solution. Say - we can take it as a particular solution, and $N(A)$ forms x_h

$$X = x_p + x_h = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ -5 \\ 1 \end{bmatrix}, \quad (\alpha, \beta) \in \mathbb{R}$$

(ii)

$$A \cdot x_0 = b, \quad A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

$$\Rightarrow 3 \begin{bmatrix} 1 \\ a_1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ a_2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ a_4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \\ 4 \end{bmatrix}$$

$$\Rightarrow 2 \begin{bmatrix} 1 \\ a_4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \\ 4 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ a_1 \\ -3 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ a_4 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ a_4 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 3 \\ 4 \end{bmatrix}$$

we know row operation won't change the linear relationship between columns of A .

From U , we can find $a_3 = 2a_1 + 3a_2$, $a_5 = -a_1 - 2a_2 + 5a_4$

$$a_3 = 2 \begin{bmatrix} 2 \\ 1 \\ -3 \\ -2 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 3 \\ -1 \end{bmatrix}$$

$$a_5 = - \begin{bmatrix} 2 \\ 1 \\ -3 \\ -2 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 2 \\ 3 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} -2 \\ -1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -10 \\ -10 \\ 12 \\ 20 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 1 & -2 & -10 \\ 1 & 2 & 8 & -1 & -10 \\ -3 & 3 & 3 & 3 & 12 \\ -2 & 1 & -1 & 4 & 20 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5$

~~✗~~

17. Determine the kernel and range of each of the following linear operators on $\mathbb{R}^3$:

(a) $L(\mathbf{x}) = (x_3, x_2, x_1)^T$ (b) $L(\mathbf{x}) = (x_1, x_2, 0)^T$

(c) $L(\mathbf{x}) = (x_1, x_1, x_1)^T$

(a) $L(\mathbf{x}) = (x_3, x_2, x_1)^T$

rewrite $L(\mathbf{x}) = A\mathbf{x}$, $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$\text{Range}(L) = C(A)$, $\text{Kernel}(L) = N(A)$

$$\text{Range}(L) = C(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\} = \mathbb{R}^3$$

Since $\text{rank}(A) = 3$ obviously, and Dimension of $N(A)$ is 0 by rank theorem. Say, $N(A) = \{0\}$.

$$\text{Kernel}(L) = N(A) = \{0\}$$

(b)

$$\text{rewrite } L(x) = Ax, \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(A) = 2$$

$$\text{Range}(L) = C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \#$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

x_3 is free variable, set $x_3 = \alpha$

$$x_1 = 0, \quad x_2 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad N(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{kernel}(L) = N(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \#$$

$$(c) \quad L(x) = Ax, \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \text{rank}(A) = 1$$

$$\text{Range}(L) = C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\} \#$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad x_2 \text{ and } x_3 \text{ are free variables}$$

$$x_2 = \alpha, \quad x_3 = \beta, \quad x_1 = 0$$

$$\vec{x} = \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad N(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{kernel}(L) = N(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \#$$

18. Let $E = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $F = \{\mathbf{b}_1, \mathbf{b}_2\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

and

$$\mathbf{b}_1 = (1, -1)^T, \quad \mathbf{b}_2 = (2, -1)^T$$

For each of the following linear transformations L from $\mathbb{R}^3$ into $\mathbb{R}^2$, find the matrix representing L with respect to the ordered bases E and F :

(a) $L(\mathbf{x}) = (x_3, x_1)^T$

(b) $L(\mathbf{x}) = (x_1 + x_2, x_1 - x_3)^T$

(c) $L(\mathbf{x}) = (2x_2, -x_1)^T$

(a)

rewrite $L(\mathbf{x}) = A\mathbf{x}$, $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$A' = F^{-1} \cdot A \cdot E, \quad E = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

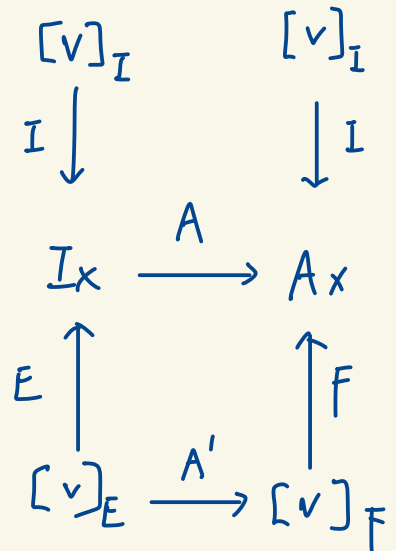
$$F = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, \quad F^{-1} = \frac{1}{\det(F)} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$A' = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -3 & 1 \\ 0 & 2 & 0 \end{bmatrix} \#$$



$$(b) \quad L(x) = Ax, \quad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$$A' = F^{-1} \cdot A \cdot E$$

$$= \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -1 & 2 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & -3 & 4 \\ 3 & 3 & -2 \end{bmatrix}^*$$

$$(c) \quad L(x) = Ax, \quad A = \begin{bmatrix} 0 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$A' = F^{-1} \cdot A \cdot E$$

$$= \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 & 0 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 3 \end{bmatrix}^*$$

20. Let V and W be vector spaces with ordered bases E and F , respectively. If $L: V \rightarrow W$ is a linear transformation

and A is the matrix representing L relative to E and F , show that

(a) $v \in \ker(L)$ if and only if $[v]_E \in N(A)$.

(b) $w \in L(V)$ if and only if $[w]_F$ is in the column space of A .

a)

$$[L(E[x]_E)]_F = [L(v)]_F = A \cdot [x]_E = [x]_F$$

i. $L(v) = 0, v \in \ker(L)$,

$$v = E[v]_E \Rightarrow L(E[v]_E) = 0$$

$$\Rightarrow [L(v)]_F = 0 \quad \left(\begin{array}{l} F^{-1} \cdot L(E[v]_E) = [L(v)]_F \\ \text{When } F \text{ is invertible} \end{array} \right)$$

As we can see, $[v]_E \in N(A)$

ii. $A[v]_E = 0, v \in N(A)$

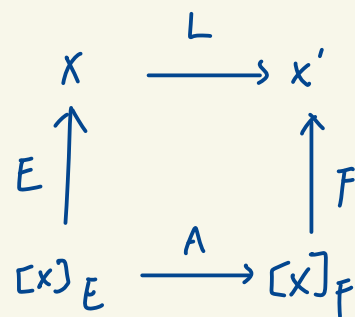
$$\Rightarrow [L(v)]_F = 0, \quad \left(F^{-1} \cdot L(E[v]_E) = 0 \right)$$

$$\Rightarrow L(v) = 0$$

Same, $v \in \ker(L)$ ✕

V

W



(Normally in $\mathbb{R}^n$ space)
 p.s. E and F are change
 of-coordinate matrices
 , so they are independent
 and invertible.

(b) i.

$$w \in L(V), \quad \forall v \in V, \quad L(v) = w$$

$$\Rightarrow [L(v)]_F = A \cdot [v]_E, \quad (L(v) = F \cdot A \cdot [v]_E)$$

$$\Rightarrow [w]_F = A \cdot [v]_E \Rightarrow \begin{bmatrix} 1 \\ w \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ a_1 & a_2 & a_3 & \dots & a_n \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ v \\ 1 \end{bmatrix}$$

From this equation, we know $A \cdot [v]_E$ can be seen as the linear combination as column vectors of A , so the result must be in the $C(A)$. Hence, $[w]_F \in C(A)$ ✖

ii. $[w]_F \in C(A)$, and exists a vector satisfied $[w]_F = A \cdot \begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix}$
rewrite $\begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix} = [v]_E$

$$[w]_F = A \cdot [v]_E$$

$$\Rightarrow F \cdot [w]_F = F \cdot A \cdot [v]_E$$

$$\Rightarrow W = F \cdot A \cdot [v]_E$$

$$\Rightarrow [w]_F = A \cdot [v]_E$$

$$\Rightarrow [w]_F = [L(v)]_F$$

Since $v \in V$ - and $w \in L(V)$ ✖

4. Let L be the linear operator mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ defined by $L(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & -2 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

and let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Find the transition matrix V corresponding to a change of basis from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, and use it to determine the matrix B representing L with respect to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

$$V \cdot [\mathbf{x}]_V = E [\mathbf{x}]_E, \quad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

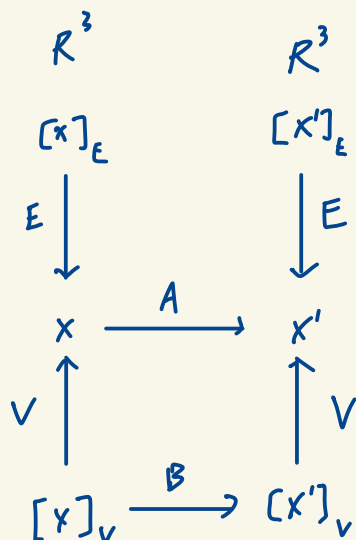
$$\Rightarrow E^{-1} V \cdot [\mathbf{x}]_V = [\mathbf{x}]_E$$

$$\Rightarrow V [\mathbf{x}]_V = [\mathbf{x}]_E$$

$$\text{transition matrix} = V = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

From the graph above, $B = V^{-1} A \cdot V$

$$\begin{aligned} [V | I] &= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 3 & -2 & -1 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 3 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -3 \\ 0 & 1 & 0 & -2 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 & 3 \end{array} \right] \Rightarrow [I | V^{-1}] \end{aligned}$$



$$B = V^{-1} \cdot A \cdot V$$

$$= \begin{bmatrix} 3 & 7 & -3 \\ -1 & 1 & 2 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & -2 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 5 \\ -1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -8 & 6 \\ -1 & 3 & -4 \\ 2 & 7 & -4 \end{bmatrix} \#$$

Chapter 3.2

a.

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rank of 6 * 6 matrix A = 6
rank of 8 * 6 matrix B = 6
rank of 5 * 6 matrix C = 5
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%(a) Explain 1 : As we know, rank can be the number of indenpdent vectors or dimensions
% in a matrix, and we know rank of column equals to rank of row, so basically the rank
% numbers r wouldn't exceed row m or column n in rank theroem.
% For example, a matrix A with m by n, we assume  $m > n$  and  $m \geq r > n$ . By rank theroem,
%  $m \geq r$  and  $n \geq r$ . Now we find a contridiction with  $m > r > n$  and  $n \geq r$ ,
% and this matrix actually doesn't have enough column
% vector to form r independent vectors, so this assumption is a
% contridiction and need to have  $r \leq n$  when  $m > n$ . With  $m > n$  and  $r \leq n$ , we can
% conclude that  $r \leq \min(m, n)$ . By symmetry, let  $m < n$  then we
% can still obtain the same result.
% (  $r \leq m$  &  $m < n \implies r \leq \min(m, n)$  )
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```
%(a) Explain 2 : For random A(m by n), we usually would expect the rank
% equals to  $\min(m, n)$ . The rank number can be seen as the number of
% indepent vector(s) for row or column vector(s). For n Column vectors in
%  $R^m$  space( $m \leq n$ ), the possibility is very low for the vectors to lay on the
% same subspace(linear dependence) in common. In other words, it is more
% difficult to generate depedent vectors compared to indepent vectors, the
% probability that any new vector falls into the subspace spanned by the
% previous vectors (linear dependence) is nearly zero,
% so we can expect  $r = m$ . By symmetry( $n \leq m$ ), we get  $r = n$ .
% In summary,  $r = \min(m, n)$  in random matrices for most of the cases.
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b.

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rank of 10 * 7 matrix A = 7
A has full rank
rank of 8 * 12 matrix B = 8
B has full rank
rank of 10 * 15 matrix C = 10
C has full rank
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c,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{1 \times n}, \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}_{1 \times m}, \quad \forall x_i \in \mathbb{R} \\ \forall y_i \in \mathbb{R}$$

$$A = X \cdot Y^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} [y_1 \ y_2 \ \dots \ y_m] \\ = \begin{bmatrix} y_1 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} & y_2 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} & \dots & y_m \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \end{bmatrix}$$

As we can see, the column vector(s) of A are composed by vector x. Every column is the multiple of x vector, which means they share the same direction and linear dependent each to each other, so we can expect $\text{rank}(A) = 1$.

45	35	5	35	25	25
63	49	7	49	35	35
81	63	9	63	45	45
54	42	6	42	30	30
54	42	6	42	30	30
36	28	4	28	20	20
45	35	5	35	25	25
63	49	7	49	35	35

the rank of A = 1

d.

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{12} & x_{22} \\ x_{13} & x_{23} \\ \vdots & \vdots \\ x_{16} & x_{26} \end{bmatrix}_{8 \times 2} = \begin{bmatrix} | & | \\ \vec{x}_a & \vec{x}_b \\ | & | \end{bmatrix}, \quad Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{12} & y_{22} \\ \vdots & \vdots \\ y_{16} & y_{26} \end{bmatrix}_{6 \times 2}$$

$$A = X \cdot Y' = \begin{bmatrix} | & | \\ \vec{x}_a & \vec{x}_b \\ | & | \end{bmatrix} \begin{bmatrix} y_{11} & y_{12} & \dots & y_{16} \\ y_{12} & y_{22} & \dots & y_{26} \end{bmatrix}$$

$$= \left[y_{11} \begin{bmatrix} | \\ \vec{x}_a \\ | \end{bmatrix} + y_{12} \begin{bmatrix} | \\ \vec{x}_b \\ | \end{bmatrix}, y_{12} \begin{bmatrix} | \\ \vec{x}_a \\ | \end{bmatrix} + y_{22} \begin{bmatrix} | \\ \vec{x}_b \\ | \end{bmatrix}, y_{13} \begin{bmatrix} | \\ \vec{x}_a \\ | \end{bmatrix} + y_{23} \begin{bmatrix} | \\ \vec{x}_b \\ | \end{bmatrix}, \dots, y_{16} \begin{bmatrix} | \\ \vec{x}_a \\ | \end{bmatrix} + y_{26} \begin{bmatrix} | \\ \vec{x}_b \\ | \end{bmatrix} \right]$$

Obviously, every column of A is the linear combination of column vectors of X , which can form a plane if $\vec{x}_a$ and $\vec{x}_b$ are linearly independent. In random non-integer matrix, the possibility that $\vec{x}_a$ and $\vec{x}_b$ are linearly dependent is extremely small, so we can expect $\text{rank}(X) = 2$, same for $\text{rank}(Y) = 2$.

The column space of A would form a line when $\text{rank}(Y) = 1$.
($\dim(C(A)) = 1$)

Assume $\text{rank}(Y) = 1$ and $\begin{bmatrix} y_{12} \\ y_{22} \end{bmatrix} = c \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}$, $c \in \mathbb{R}$

(linear dependent for each pair)

The first and second column of A becomes $x_{11} \cdot \begin{bmatrix} 1 \\ x_a \\ 1 \end{bmatrix} + x_{12} \begin{bmatrix} 1 \\ x_b \\ 1 \end{bmatrix}$

and $c \left(x_{11} \cdot \begin{bmatrix} 1 \\ x_a \\ 1 \end{bmatrix} + x_{12} \begin{bmatrix} 1 \\ x_b \\ 1 \end{bmatrix} \right)$,

We can find they are linear dependent, so $\text{rank}(A) = 2$ needs $\text{rank}(Y) = 2$. However, the possibility that $\text{rank}(Y) < 2$ is extremely small as well.

Hence, we can expect the column space of A is confined in a plane spanned by $\bar{x}_a$ and $\bar{x}_b$ and $\text{rank}(A) = 2$
($\dim(C(A)) = 2$)

```
>> HW5_2d
```

0.0203	0.2208	0.0476	0.0772	0.2878	0.3755
0.1141	0.2642	0.1066	0.1919	0.5154	0.5879
0.0497	0.2979	0.0765	0.1288	0.4306	0.5409
0.0408	0.3699	0.0835	0.1367	0.4948	0.6393
0.0571	0.5341	0.1196	0.1955	0.7113	0.9205
0.0589	0.4127	0.1005	0.1674	0.5777	0.7341
0.0781	0.2491	0.0842	0.1482	0.4297	0.5087
0.0852	0.4831	0.1267	0.2140	0.7072	0.8844

the rank of $A = 2$

Chapter 4.1

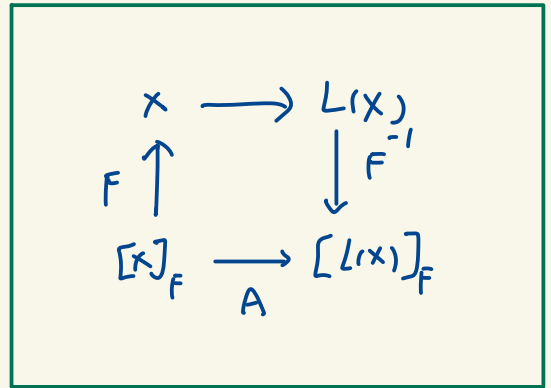
$$1a) L(W) = \left[\begin{array}{c|c|c|c} | & | & | & | \\ L(w_1) & L(w_2) & L(w_3) & L(w_4) \\ | & | & | & | \end{array} \right]$$

$$L(w_1) = \left[\begin{array}{c|c|c|c} | & | & | & | \\ w_1 & w_2 & w_3 & w_4 \\ | & | & | & | \end{array} \right] \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = W \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$L(w_2) = W \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$L(w_3) = W \begin{bmatrix} 5 \\ 1 \\ 3 \\ 2 \end{bmatrix}$$

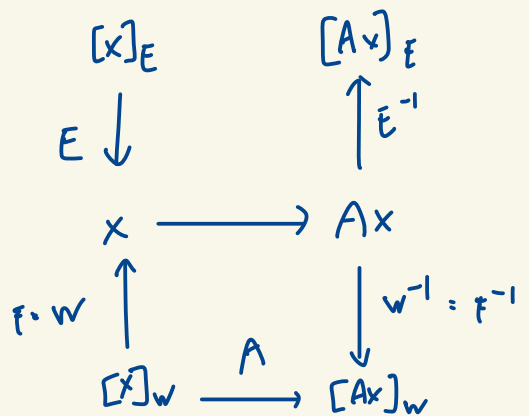
$$L(w_4) = W \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$



$$\left[W \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad W \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad W \begin{bmatrix} 5 \\ 1 \\ 3 \\ 2 \end{bmatrix} \quad W \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \right]$$

$$= \left[\begin{array}{c|c|c|c} | & | & | & | \\ w_1 & w_2 & w_3 & w_4 \\ | & | & | & | \end{array} \right] \begin{bmatrix} 0 & 0 & 5 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow [L]_F = A$$

$$A = \begin{bmatrix} 0 & 0 & 5 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$



$$(b) \quad y = W^{-1}x = [x]_W$$

$$(c) \quad z = A[x]_W$$

$$(d) \quad Ax = L(x) = W \cdot z$$

Question a

$$L(W) = WA, \quad A =$$

$$\begin{bmatrix} 0 & 0 & 5 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Question b

$$y =$$

$$\begin{bmatrix} -1 \\ -1 \\ -1 \\ 4 \end{bmatrix}$$

Question c

$$z =$$

$$\begin{bmatrix} -1 \\ 3 \\ 8 \\ 14 \end{bmatrix}$$

Question d

$$L(x) =$$

$$\begin{bmatrix} 24 \\ 25 \\ 22 \\ 14 \end{bmatrix}$$